



Mixture-of-Prefetchers

A compact L2-cache router tuned by OpenEvolve



Trace behavior shifts

The stronger L2C prefetcher changes by workload, so one fixed expert leaves performance unused.



Route cheaply

MoP observes short epoch counters, then chooses how much budget each expert receives.



Let search tune policy

OpenEvolve adjusts compact routing constants while the simulator and evaluation protocol stay fixed.



Show the evidence

Every plot uses prefetcher off as 1.000x and labels the best expert as a reference line.



What We Built On Athena



Contribution over Athena

Athena provides the simulator, cache hierarchy, L2C prefetchers, and existing comparison baselines. Our work adds a two-prefetcher routing layer, epoch-level expert counters, fixed train and heldout trace protocol, an OpenEvolve evaluator, and reproducible ledgers and plots.



Expert pair: Athena MLOP + SPP+PPF at L2C.



MoP-V1: manual one-epoch probe router.



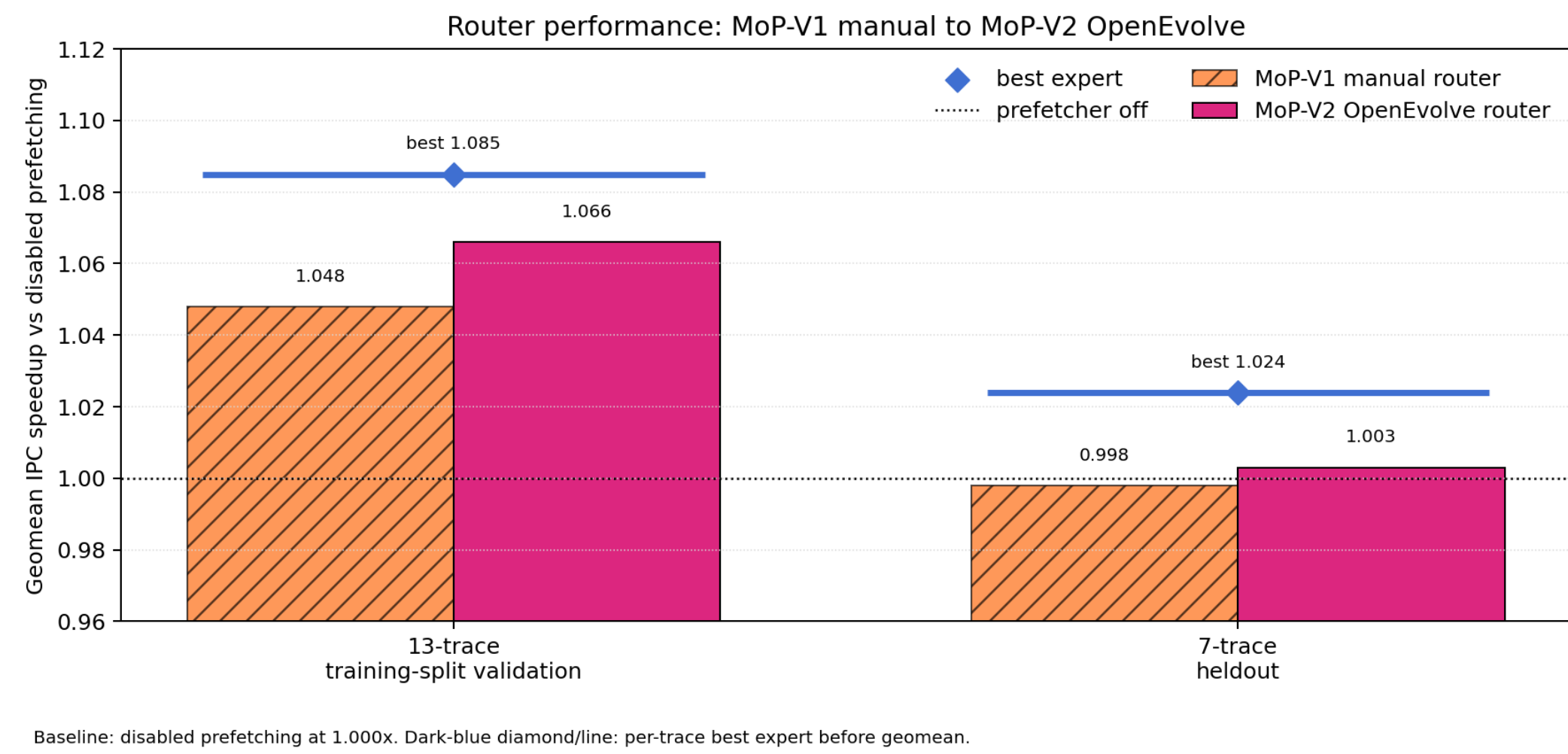
MoP-V2: OpenEvolve-tuned MoP-V1 with evolved budget, score weights, and close-score tie margin.



OpenEvolve could change only a compact policy dictionary. Trace split, baselines, parser, metrics, expert pair, cache level, and simulator internals stayed fixed.



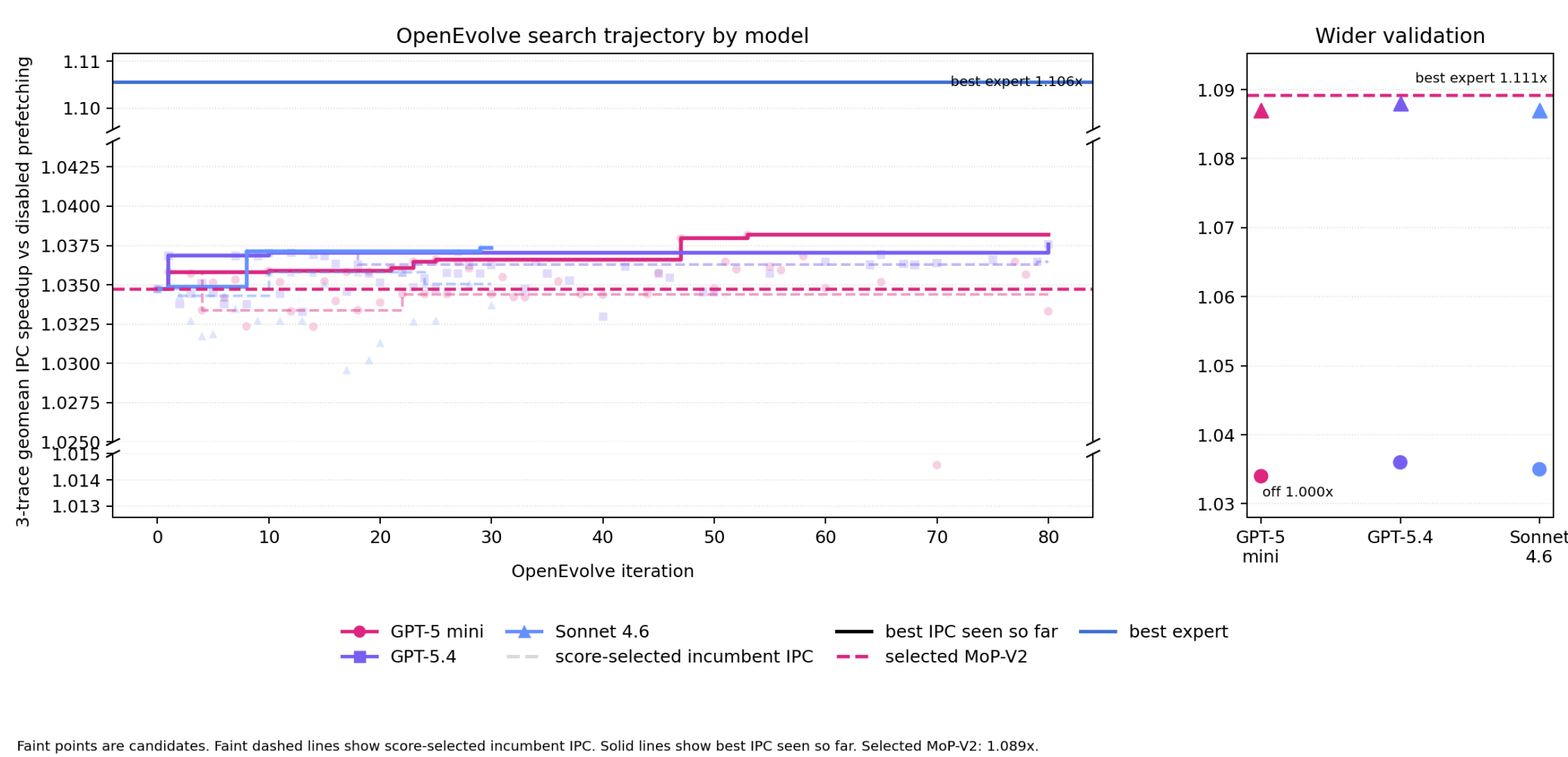
Before and after OpenEvolve



Caption: bars compare disabled prefetching, each single expert, the manual MoP-V1 router, the OpenEvolve-tuned MoP-V2 router, and the best expert reference on the same heldout and training-validation surfaces.



Search trajectory by model



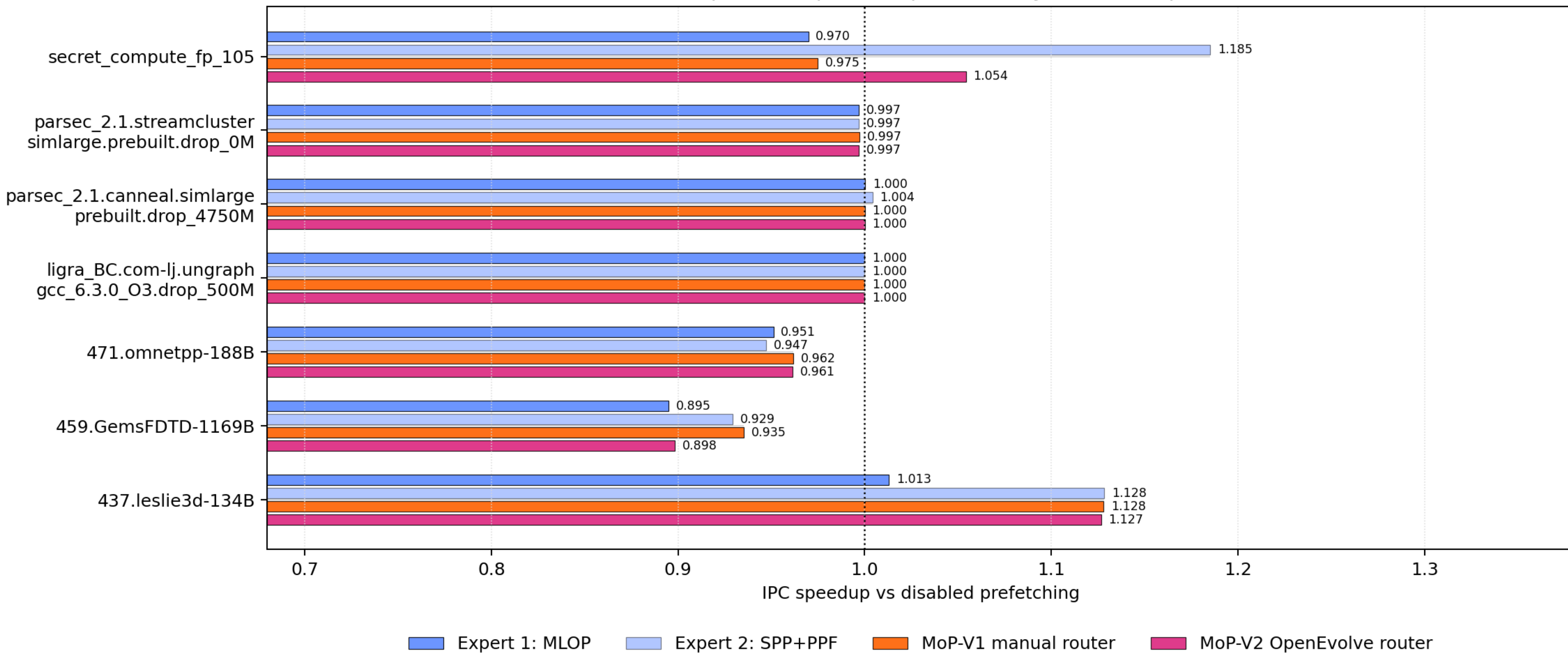
Caption: faint points are generated candidates. Solid traces show best-so-far weighted evaluator score, which combines speedup, percent of best expert, weaker-expert coverage, and tail-loss penalties.



Heldout trace profile

Caption: each row is a heldout trace. Bars use disabled prefetching as 1.000x. MLOP and SPP+PPF sit above the MoP bars so the reader can see when the two experts disagree and where the router lands.

Heldout trace profile: expert complementarity and router placement



Each trace row is ordered top-to-bottom: Expert 1, Expert 2, MoP-V1, and MoP-V2. Black dotted line is disabled prefetching at 1.000x.



Takeaway

For MLOP + SPP+PPF, MoP-V2 improves over the manual MoP-V1 router on training-split validation, stays positive on heldout by the disabled-prefetching baseline, and keeps the best expert as a separate oracle-style reference.



Auditability

Candidate ledgers record 374 attempts, including 306 valid scored rows and 68 fail-closed rows. The detailed naming map, raw result paths, CI policy, and rebuild commands live in report/writing_logistics.md.